

# Convex Optimization

## Home Work - 2

### Exercise 1

1) (P) : 
$$\begin{cases} \min_x c^T x \\ \text{s.t. } Ax - b = 0 \\ -x \leq 0 \end{cases}$$

the Lagrangian function  $L(x, \lambda, v) = c^T x + \lambda^T (-x) + v^T (Ax - b)$   

$$= (c + A^T v - \lambda)^T x - v^T b$$
  

$$= (c - \lambda + A^T v)^T x - b^T v$$

→ the Lagrange dual function  $g$  :  $g(\lambda, v) = \inf_{x \in D} L(x, \lambda, v)$  let  $D = \{x \in \mathbb{R}^n \mid Ax - b = 0, -x \leq 0\}$

$$g(\lambda, v) = \inf_{x \in D} (c - \lambda + A^T v)^T x - b^T v$$

$$= -b^T v + \inf_{x \in D} (c - \lambda + A^T v)^T x$$

$$= \begin{cases} -b^T v & \text{if } c - \lambda + A^T v = 0 \\ -\infty & \text{otherwise} \end{cases}$$

then the Lagrange dual problem can be written as :  $\begin{cases} \max_{\lambda, v} g(\lambda, v) \\ \lambda \geq 0 \end{cases} \Leftrightarrow \begin{cases} \max_v -b^T v \\ A^T v - \lambda + c = 0 \\ \lambda \geq 0 \end{cases}$

$$\Leftrightarrow \begin{cases} \max_v -b^T v \\ A^T v + c \geq 0 \end{cases}$$

2)  $D = \begin{cases} \max_y b^T y \\ A^T y \leq c \end{cases} \Leftrightarrow \begin{cases} \min -b^T y \\ A^T y - c \leq 0 \end{cases}$

→ the Lagrangian function :

$$L(y, \lambda) = -b^T y + \lambda^T (A^T y - c) = (A\lambda - b)^T y - c^T \lambda$$

let  $D = \{y \in \mathbb{R}^n \mid A^T y - c \leq 0\}$

$$g(\lambda) = \inf_{y \in D} L(y, \lambda) = \inf_{y \in D} (A\lambda - b)^T y - c^T \lambda$$

$$(A\lambda - b)^T y - c^T \lambda = \begin{cases} -c^T \lambda & \text{if } A\lambda - b = 0 \\ -\infty & \text{otherwise} \end{cases}$$

the dual problem can be written  $\begin{cases} \max & -c^T \lambda \\ A\lambda - b = 0 \\ \lambda \geq 0 \end{cases} \Leftrightarrow \begin{cases} \min & c^T \lambda \\ A\lambda = b \\ \lambda \geq 0 \end{cases} = (P)$

3)  $\begin{cases} \min_{x,y} & c^T x - b^T y \\ \text{s.t.} & Ax = b \\ & x \geq 0 \\ & A^T y \leq c \end{cases} \Leftrightarrow \begin{cases} \min_{x,y} & c^T x - b^T y \\ \text{s.t.} & Ax - b = 0 \\ & -x \leq 0 \\ & A^T y - c \leq 0 \end{cases}$

$$L((x,y), \lambda_1, \lambda_2, v) = c^T x - b^T y + \lambda_1^T (-x) + \lambda_2^T (A^T y - c) + v^T (Ax - b)$$

$$= (c^T - \lambda_1^T + A^T v)^T x + (A \lambda_2 - b)^T y - (\lambda_2^T c + v^T b)$$

$$D = \{(x,y) \mid Ax = b; x \geq 0; A^T y \leq c\}$$

$$g(\lambda_1, \lambda_2, v) = \inf_{(x,y) \in D} L((x,y), \lambda_1, \lambda_2, v) = \begin{cases} \inf_y (A \lambda_2 - b)^T y - (\lambda_2^T c + v^T b) & \text{if } c - \lambda_1 + A^T v = 0 \text{ and } A \lambda_2 - b = 0 \\ -\infty & \text{otherwise} \end{cases}$$

$$= \begin{cases} -(\lambda_2^T c + v^T b) & \text{if } c - \lambda_1 + A^T v = 0 \text{ and } A \lambda_2 - b = 0 \\ -\infty & \text{otherwise} \end{cases}$$

the dual problem can be written as

$$\begin{cases} \max_{\lambda_1, \lambda_2, v} & -c^T \lambda_2 - b^T v \\ \text{s.t.} & A \lambda_2 - b = 0 \\ & c - \lambda_1 + A^T v = 0 \\ & -\lambda_1 \leq 0 \\ & -\lambda_2 \leq 0 \end{cases}$$

$$\Leftrightarrow \begin{cases} \max_{\lambda_2, v} & -c^T \lambda_2 - b^T v \\ \text{s.t.} & A \lambda_2 - b = 0 \\ & -(c + A^T v) \leq 0 \\ & -\lambda_2 \leq 0 \end{cases}$$

$$(v = -u) \Leftrightarrow \begin{cases} \max_{\lambda_2, v} & -c^T \lambda_2 + b^T v \\ \text{s.t.} & A \lambda_2 - b = 0 \\ & -A^T v + c \geq 0 \\ & \lambda_2 \geq 0 \end{cases}$$



$$\Leftrightarrow \begin{cases} \min_{\lambda_2, V} c^T \lambda_2 - b^T V \\ A \lambda_2 - b = 0 \\ A^T V - c \leq 0 \\ -\lambda_2 \leq 0 \end{cases} \Leftrightarrow \begin{cases} \min_y c^T x - b^T y \\ \text{s.t.} \\ A x - b = 0 \\ -x \leq 0 \\ A^T y - c \leq 0 \end{cases}$$

$\Rightarrow$  the problem is self dual because its dual problem is the problem itself

$$D = \{ (x, y) \mid Ax = b, x \geq 0, A^T y \leq c \} = \{ x \mid Ax = b, x \geq 0 \} \cup \{ y \mid A^T y \leq c \}$$

$$\text{so } \begin{cases} \min_{x, y} c^T x - b^T y \\ \text{s.t.} \\ Ax = b \\ x \geq 0 \\ A^T y \leq c \end{cases} \Leftrightarrow \begin{cases} \min_x c^T x \\ \text{s.t.} \\ Ax = b \\ -x \leq 0 \end{cases} + \begin{cases} \min_y -b^T y \\ \text{s.t.} \\ A^T y - c \leq 0 \end{cases}$$

$$(H) \Leftrightarrow \{ (P) \} + \begin{cases} \max_y b^T y \\ \text{s.t.} \\ A^T y - c \leq 0 \end{cases} \Leftrightarrow (P) + (D)$$

if  $(x^*, y^*)$  is a solution of the (H) so  $x^*$  is an optimal solution for P and  $y^*$  is an optimal

solution for (D)

$\rightarrow$  the problem (P) is strictly feasible since it is a convex problem (the objective function and the constraint are linear), its optimal solution is constant qualification. Hence, it is strictly feasible for (P)

$\rightarrow$  (D) is the dual problem of (P). Thus we have  $p^* = d^*$  ~~and~~  $p^* = d^* \Leftrightarrow c^T x^* - b^T y^* = 0$

So the optimal value of (H) is 0

Exercises

## Exercise 2

We have for a given  $A \in \mathbb{R}^{h \times d}$  and  $b \in \mathbb{R}^h$ , (P):  $\min_x \|Ax - b\|_2^2 + \|x\|_1$ .

$$1) \|x\|_1 = \sum_{i=1}^d |x_i|$$

$$f^*(y) = \sup_{x \in \mathbb{R}^d} (y^T x - \|x\|_1) = \sup_{x \in \mathbb{R}^d} \left( y^T \sum_{i=1}^d x_i e_i - \sum_{i=1}^d |x_i| \right)$$

$$\text{if } |y_i| \leq 1: x_i y_i - |x_i| \leq |x_i| - |x_i| = 0$$

$$\Rightarrow \sum_{i=1}^d x_i y_i - \|x\|_1 \leq 0$$

$$\Rightarrow y^T x - \|x\|_1 \leq 0$$

Dual here

$$\|z\|_\infty = \sup \{ z^T x \mid \|x\|_1 \leq 1 \}$$

$$= \|z\|_\infty$$

$$\Rightarrow f^*(y) = 0$$

if  $\|y\|_\infty > 1 \exists z$  s.t.  $\|z\|_1 < 1$  and  $y^T z > 1$  let  $x = tz$  and  $t \rightarrow \infty$

we have  $y^T x - \|x\|_1 = t(y^T z - \|z\|_1) \rightarrow \infty$

we shows that  $f^*(y) = \infty$ . conversely if  $\|y\|_\infty \leq 1$  we have  $y^T x \leq \|x\|_1$

for all  $x$ . we implies for all  $x, y^T x - \|x\|_1 \leq 0$  thus  $x=0$  is the max that maximizes  $y^T x - \|x\|_1$  is max value 0

$$f^*(y) = \begin{cases} 0 & \|y\|_\infty \leq 1 \\ \infty & \text{otherwise} \end{cases}$$

$$f^*(y) = \begin{cases} 0 & \|y\|_\infty \leq 1 \\ \infty & \text{otherwise} \end{cases}$$

$$2) \text{ RLS: } \min_x \|Ax - b\|_2^2 + \|x\|_1 \Leftrightarrow \min_{\substack{y \\ Ax - b - y = 0}} \|y\|_2^2 + \|x\|_1$$

$$L(x, y, v) = \|y\|_2^2 + \|x\|_1 + v^T (Ax - y - b)$$

$$0 = h(x, y) \mid Ax - y - b = 0$$

$$g(v) = \inf_{x, y \in \mathbb{R}^d} \{ \|y\|_2^2 + \|x\|_1 + v^T (Ax - y - b) \}$$

$$= \inf_{x, y \in \mathbb{R}^d} \{ \|x\|_1 + v^T Ax + y^T y - v^T y - v^T b \}$$

$$= \inf_{x \in \mathbb{R}^d} \{ \|x\|_1 + v^T Ax + y^T y - v^T y - v^T b \}$$



$$g(v) = \inf_n \{ \|v\|_1 + v^T A x \} + \inf_y \{ \underbrace{y^T y - v^T y}_{h(y)} \} - b^T v$$

$h(y) = y^T y - v^T y$  is differentiable

$$\nabla h = 2y - v = 0 \Rightarrow y = \frac{v}{2}$$

$$\inf_y \{ h(y) \} = \frac{1}{4} \|v\|_2^2 - \frac{1}{2} \|v\|_2^2 = -\frac{1}{4} v^T v$$

$$\inf_n \{ \|x\|_1 + v^T A x \} = -\sup_n \{ (-A^T v)^T x - \|x\|_1 \} \stackrel{q.s.}{=} \begin{cases} 0 & \text{if } \|A^T v\|_\infty \leq 1 \\ \infty & \text{otherwise} \end{cases}$$

$$g(v) = \begin{cases} -\frac{v^T v}{4} - b^T v & \text{if } \|A^T v\|_\infty \leq 1 \\ -\infty & \text{otherwise} \end{cases}$$

dual problem is

$$\begin{cases} \max_v & -\frac{\|v\|_2^2}{4} - v^T b \\ & \|A^T v\|_\infty \leq 1 \end{cases}$$

$$\Leftrightarrow \begin{cases} \min & \frac{v^T v}{4} + b^T v \\ \text{s.t.} & \|A^T v\|_\infty \leq 1 \end{cases}$$

# Exercice 3

$$1) \min_w \frac{1}{n} \sum_{i=1}^n L(w^T x_i, y_i) + \frac{\sigma}{2} \|w\|_2^2 \quad (\text{sep})$$

$$\Leftrightarrow \min_w \frac{1}{n} \sum_{i=1}^n \max \{0, 1 - y_i (w^T x_i)\} + \frac{\sigma}{2} \|w\|_2^2$$

~~Noter:~~

$$\max \{0, 1 - y_i (w^T x_i)\} = z_i \quad \forall i$$

$$\text{if } z_i = 0 \Rightarrow 1 - y_i (w^T x_i) \leq 0 = z_i$$

$$\text{if } z_i = 1 \rightarrow 1 - y_i (w^T x_i) = 1 = z_i$$

$$\Rightarrow \max \{0, 1 - y_i (w^T x_i)\} = z_i \quad \forall i$$

$$\Leftrightarrow \begin{cases} 1 - y_i (w^T x_i) \leq z_i \\ z_i \geq 0 \end{cases}$$

$$\Rightarrow \min_w \frac{1}{n} \sum_{i=1}^n L(w^T x_i, y_i) + \frac{\sigma}{2} \|w\|_2^2$$

$$\Leftrightarrow \begin{cases} \min_{w, z_i} \frac{1}{n} \sum_{i=1}^n z_i + \frac{\sigma}{2} \|w\|_2^2 \\ \text{s.t. } 1 - y_i (w^T x_i) \leq z_i \\ z_i \geq 0 \end{cases}$$

$$\Leftrightarrow \begin{cases} \min_{w, z} \frac{1}{n} \sum_{i=1}^n z_i + \frac{\sigma}{2} \|w\|_2^2 \\ 1 - y_i (w^T x_i) \leq z_i \quad \forall i \in [1, n] \\ z_i \geq 0 \end{cases} \rightarrow (\text{sep } 2)$$

so (sep 2) solves (sep 1)

$$2- L(w, \lambda, \pi) = \frac{1}{n} \sum_{i=1}^n z_i + \frac{\sigma}{2} \|w\|_2^2 - \pi^T z + \sum_{i=1}^n \lambda_i (1 - z_i - y_i (w^T x_i))$$

$$= \left[ \frac{1}{n} - \pi - \lambda \right]^T z + \frac{\sigma}{2} \|w\|_2^2 + \sum_{i=1}^n \lambda_i (1 - y_i (w^T x_i))$$

$$g(\lambda, \pi) = \inf_{z, w} L(w, \lambda, \pi) \quad ; \quad D = \{ (z, w) \mid z \leq 0, 1 - y_i (w^T x_i) \leq z_i \text{ (oté)} \}$$

$$g(\lambda, \pi) = \inf_w \left\{ \inf_z \left\{ \left[ \frac{1}{n} - \pi - \lambda \right]^T z + \frac{\sigma}{2} \|w\|_2^2 + \sum_{i=1}^n \lambda_i (1 - y_i (w^T x_i)) \right\} \right\}$$

$$= \inf_w \left\{ \frac{1}{2} w^T w - \sum_{i=1}^n \lambda_i y_i (w^T x_i) + \inf_z \left\{ \left[ \frac{1}{n} - \pi - \lambda \right]^T z \right\} + 1^T \lambda \right\}$$

$$\text{let } h(w) = \frac{1}{2} w^T w - \sum_{i=1}^n \lambda_i y_i (x_i^T w)$$

$$\frac{\partial h}{\partial w} = w - \sum_{i=1}^n \lambda_i y_i x_i = 0 \Rightarrow w = \sum_{i=1}^n \lambda_i y_i x_i$$

$$\begin{aligned} \Rightarrow \inf_w \left\{ \frac{1}{2} w^T w - \sum_{i=1}^n \lambda_i y_i x_i^T w \right\} &= \frac{1}{2} \left\| \sum_{i=1}^n \lambda_i y_i x_i \right\|_2^2 - \left\| \sum_{i=1}^n \lambda_i y_i x_i \right\|_2^2 \\ &= -\frac{1}{2} \left\| \sum_{i=1}^n \lambda_i y_i x_i \right\|_2^2 \end{aligned}$$

$$\text{let } g(\lambda) = \left( \frac{1}{n^2} - \pi - \lambda \right)^T \lambda$$

$$\inf_{\lambda} \left\{ \left( \frac{1}{n^2} - \pi - \lambda \right)^T \lambda \right\} = \begin{cases} 0 & \text{if } \frac{1}{n^2} - \pi - \lambda = 0 \\ -\infty & \text{otherwise} \end{cases}$$

$$\Rightarrow g(\lambda, \pi) = \begin{cases} -\frac{1}{2} \left\| \sum_{i=1}^n \lambda_i y_i x_i \right\|_2^2 + \pi^T \lambda & \text{if } \frac{1}{n^2} - \pi - \lambda = 0 \\ -\infty & \text{otherwise} \end{cases}$$

we can write the dual problem as:

$$\begin{cases} \max_{\lambda, \pi} & -\frac{1}{2} \left\| \sum_{i=1}^n \lambda_i y_i x_i \right\|_2^2 + \pi^T \lambda \\ \text{s.t.} & \frac{1}{n^2} - \pi - \lambda = 0 \\ & \lambda \geq 0 \\ & \pi \geq 0 \end{cases} \Leftrightarrow \begin{cases} \max_{\lambda} & -\frac{1}{2} \left\| \sum_{i=1}^n \lambda_i y_i x_i \right\|_2^2 + \pi^T \lambda \\ & \frac{1}{n^2} - \lambda = \pi \geq 0 \\ & \lambda \geq 0 \end{cases}$$

$$\Leftrightarrow \begin{cases} \max_{\lambda} & -\frac{1}{2} \left\| \sum_{i=1}^n \lambda_i y_i x_i \right\|_2^2 + \pi^T \lambda \\ & \lambda - \frac{1}{n^2} \leq 0 \\ & -\lambda \leq 0 \end{cases}$$